frq8

1. $$3\ Ag(s) + 4\ HNO_3(aq) \rightarrow 3\ AgNO_3(aq) + NO(g) + 2\ H_2O(l)$$

A student investigates the reaction between $Ag(s)$ and $HNO_3(aq)$ represented by the equation above.

a. Predict the sign of the entropy change, $\Delta S°$, for the reaction. Justify your answer.

b. Use the information in the table below to calculate the value $\Delta H_{rxn}°$, the standard enthalpy change for the reaction, in kJ/mol$_{rxn}$.

Substance	$HNO_3(aq)$	$AgNO_3(aq)$	$NO(g)$	$H_2O(l)$
Standard Heat of Formation, $\Delta H_f°$ (kJ/mol)	−207	−101	90.	−286

c. Based on your answers to parts (a) and (b), is the reaction more likely to be thermodynamically favorable at 25°C, or at 95°C? Justify your answer.

d. The student runs the reaction using a 3 to 4 mole ratio of $Ag(s)$ to $HNO_3(aq)$. Suggest a method the student can use to isolate solid $AgNO_3$ from the other products of the reaction.

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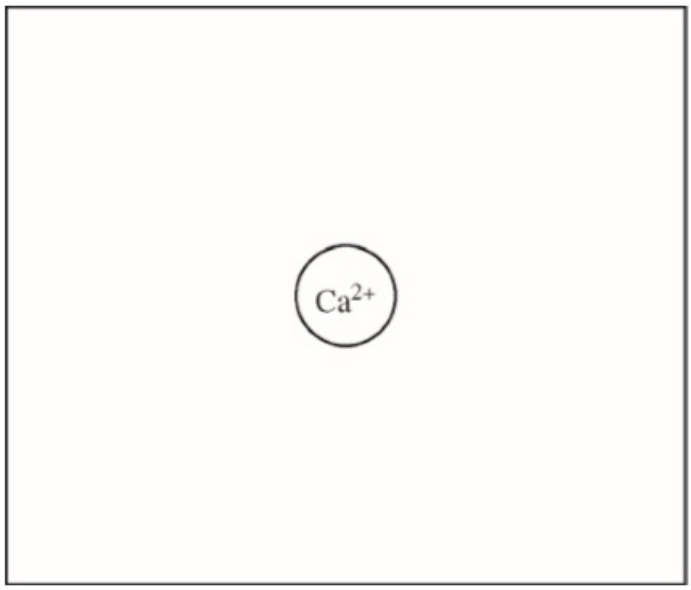
2. Answer the following questions about the solubility of $Ca(OH)_2$ ($K_{sp} = 1.3 \times 10^{-6}$) .

a. Write a balanced chemical equation for the dissolution of $Ca(OH)_2(s)$ in pure water.

b. Calculate the molar solubility of $Ca(OH)_2$ in 0.10 *M* $Ca(NO_3)_2$.

c. In the box below, complete a particle representation diagram that includes four water molecules with proper
orientation around the Ca^{2+} ion.

Represent water molecules as .



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3. For parts of the free response question that require calculations, clearly show the method used and the steps involved in arriving at your answers. You must show your work to receive credit for your answer. Examples and equations may be included in your answers where appropriate.

For parts of the free response question that require calculations, clearly show the method used and the steps involved in arriving at your answers. You must show your work to receive credit for your answer. Examples and equations may be included in your answers where appropriate.

A student is asked to prepare a buffer solution made with equimolar amounts of $CH_3NH_2(aq)$ and $CH_3NH_3Cl(s)$. The student uses $25.00\ \text{mL}$ of $0.100\ M\ CH_3NH_2(aq)$, which contains $0.00250\ \text{mol}$ of CH_3NH_2, to make the buffer.

(a) Calculate the mass of $CH_3NH_3Cl(s)$ that contains $0.00250\ \text{mol}$ of CH_3NH_3Cl.

The student has the following materials and equipment available.

• Distilled water	• Electronic balance	• $50\ \text{mL}$ beaker	• Pipets
• $0.100\ M\ CH_3NH_2(aq)$	• Weighing paper	• $10.0\ \text{mL}$ graduated cylinder	• pH meter
• Solid CH_3NH_3Cl	• $50.00\ \text{mL}$ buret	• Small spatula	

(b) The following table contains a partial procedure for making the buffer solution. Fill in steps 1 and 4 to complete the procedure using only materials and equipment selected from the choices given. (Not all materials listed will be used. Assume that all appropriate safety measures are already in place.)

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Step	Procedure
1	
2	Place the solid in the 50mL beaker.
3	Clean the buret and rinse with distilled water.
4	
5	Use the buret to add 25.00mL of $0.100M\text{CH}_3\text{NH}_2(aq)$ to the beaker.
6	Mix well.
7	Check the pH with the pH meter.

The value of K_b for $\text{CH}_3\text{NH}_2(aq)$ is 4.4×10^{-4}, and the pH of the buffer the student prepared is 10.64.

(c) The student prepares a second buffer solution. The student uses $25.00\ \text{mL}$ of $0.050\ M\ \text{CH}_3\text{NH}_2(aq)$ instead of $25.00\ \text{mL}$ of $0.100\ M\ \text{CH}_3\text{NH}_2(aq)$, and half the mass of $\text{CH}_3\text{NH}_3\text{Cl}(s)$ that was used in the first buffer. Is the pH of the second buffer greater than, less than, or equal to the pH of the first buffer? Justify your answer.

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4. Directions: For parts of the free-response question that require calculations, clearly show the method used and the steps involved in arriving at your answers. You must show your work to receive credit for your answer. Examples and equations may be included in your answers where appropriate.

Equimolar amounts of $Cl_2(g)$ and $CO(g)$ are injected into an evacuated, rigid container, where they react according to the equation below.

$$Cl_2(g) + CO(g) \rightleftarrows COCl_2(g) \qquad \Delta H^{\circ}_{rxn} = -109\,\text{kJ/mol}_{rxn}$$

(a) If 7.0 g of $CO(g)$ is consumed in the reaction with excess $Cl_2(g)$, how many moles of $COCl_2(g)$ are produced?

(b) Which element is oxidized in this reaction? Justify your answer in terms of oxidation numbers.

The partial pressures of the gases in the container are monitored at constant temperature and recorded in the table below.

Time	P_{Cl_2} (atm)	P_{CO} (atm)	P_{COCl_2} (atm)
t_0	1.0	1.0	0.0
t_1	0.78	0.78	0.22
t_2	0.62	0.62	0.38
t_3	0.51	0.51	0.49
t_4	0.45	0.45	0.55
t_5	0.42	0.42	0.58
t_6	0.41	0.41	0.59
t_7	0.40	0.40	0.60
t_8	0.40	0.40	0.60
t_9	0.40	0.40	0.60

(c) At time t_8, is the rate of the forward reaction greater than, less than, or equal to the rate of the reverse reaction? Justify your choice.

(d) At equilibrium, the container holds the most molecules of which gas, $Cl_2(g)$ or $COCl_2(g)$? Explain your answer.

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(e) A student hypothesizes that if the temperature of the container is decreased after time t_9, the mole fraction of Cl_2 in the container will increase. Do you agree or disagree with the student's hypothesis? Justify your answer.

(f) Using the data in the table, determine the value of the equilibrium constant, K_p, for the reaction represented by the equation below.

$$Cl_2(g) + CO(g) \rightleftarrows COCl_2(g)$$

(g) Using the value determined in part (f), determine the value of $K_p\prime$ for the equation shown below.

$$3\,Cl_2(g) + 3\,CO(g) \rightleftarrows 3\,COCl_2(g) \qquad K_p\prime = ?$$

(h) Which gas, $Cl_2(g)$ or $COCl_2(g)$, will deviate most from the ideal gas law at low temperature? Justify your choice.

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5. Directions: For parts of the free-response question that require calculations, clearly show the method used and the steps involved in arriving at your answers. You must show your work to receive credit for your answer. Examples and equations may be included in your answers where appropriate.

Equimolar amounts of $Cl_2(g)$ and $O_2(g)$ are injected into an evacuated, rigid container, where they react according to the equation below.

$$2\,Cl_2(g) + O_2(g) \rightleftarrows 2\,Cl_2O(g) \qquad \Delta H^\circ_{rxn} = +161\ \text{kJ/mol}_{rxn}$$

(a) If 6.4 g of $O_2(g)$ is consumed in the reaction with excess $Cl_2(g)$, how many moles of $Cl_2O(g)$ are produced?

(b) Which element is oxidized in this reaction? Justify your answer in terms of oxidation numbers.

The partial pressures of the gases in the container are monitored at constant temperature and recorded in the table below.

Time	P_{Cl_2} (atm)	P_{O_2} (atm)	P_{Cl_2O} (atm)
t_0	0.50	0.50	0.0
t_1	0.42	0.46	0.08
t_2	0.36	0.43	0.14
t_3	0.32	0.41	0.19
t_4	0.29	0.39	0.22
t_5	0.27	0.38	0.23
t_6	0.26	0.37	0.24
t_7	0.25	0.38	0.25
t_8	0.25	0.38	0.25
t_9	0.25	0.38	0.25

(c) At time t_4, is the rate of the reverse reaction greater than, less than, or equal to the rate of the forward reaction? Justify your choice.

(d) At equilibrium, the container holds fewer molecules of which gas, $Cl_2(g)$ or $O_2(g)$? Explain your answer.

(e) A student hypothesizes that if the temperature of the container is decreased after time t_9, the mole fraction of $Cl_2O(g)$ in the container will increase. Do you agree or disagree with the student's hypothesis? Justify your answer.

(f) Using the data in the table, determine the value of the equilibrium constant, K_p, for the reaction represented by

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the equation below.

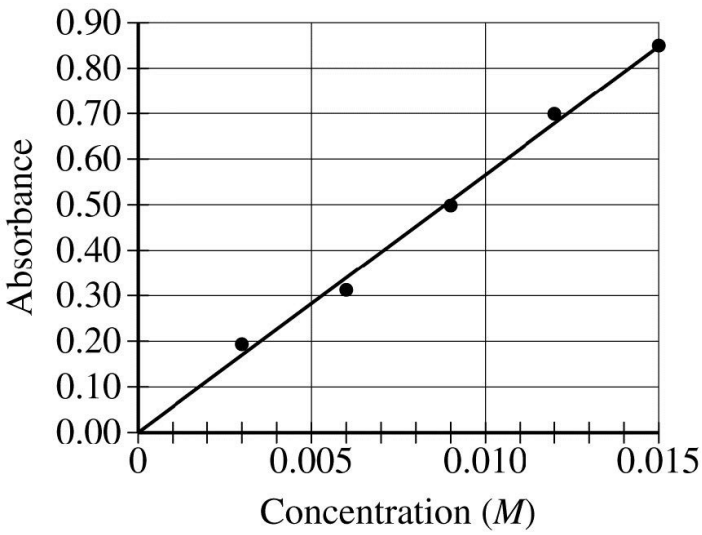
$$2\,Cl_2(g) + O_2(g) \rightleftarrows 2\,Cl_2O(g)$$

(g) Using the value determined in part (f), determine the value of $K_p\prime$ for the equation shown below.

$$4\,Cl_2(g) + 2\,O_2(g) \rightleftarrows 4\,Cl_2O(g) \qquad K_p\prime = ?$$

(h) Which gas, $Cl_2(g)$ or $O_2(g)$, will deviate most from the ideal gas law at low temperature? Justify your choice.

6. To spectrophotometrically determine the mass percent of cobalt in an ore containing cobalt and some inert materials, solutions with known $[Co^{2+}]$ are prepared and the absorbance of each of the solutions is measured at the wavelength of optimum absorbance. The data are used to create a calibration plot, shown below.



A 0.630 g sample of the ore is completely dissolved in concentrated $HNO_3(aq)$. The mixture is diluted with water to a final volume of 50.00 mL. Assume that all the cobalt in the ore sample is converted to $Co^{2+}(aq)$.

a. What is the $[Co^{2+}]$ in the solution if the absorbance of a sample of the solution is 0.74?

b. Calculate the number of moles of $Co^{2+}(aq)$ in the 50.00 mL solution.

c. Calculate the mass percent of Co in the 0.630 g sample of the ore.

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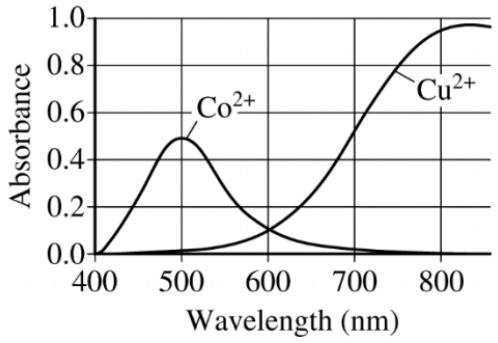
7. **Directions:** Examples and equations may be included in your responses where appropriate. For calculations, clearly show the method used and the steps involved in arriving at your answers. You must show your work to receive credit for your answer. Pay attention to significant figures.

A student has 100. mL of 0.400 M $CuSO_4(aq)$ and is asked to make 100. mL of 0.150 M $CuSO_4(aq)$ for a spectrophotometry experiment. The following laboratory equipment is available for preparing the solution: centigram balance, weighing paper, funnel, 10 mL beaker, 150 mL beaker, 50 mL graduated cylinder, 100 mL volumetric flask, 50 mL buret, and distilled water.

(a) Calculate the volume of 0.400 M $CuSO_4(aq)$ required for the preparation.

(b) Briefly describe the essential steps to most accurately prepare the 0.150 M $CuSO_4(aq)$ from the 0.400 M $CuSO_4(aq)$ using the equipment listed above.

The student plans to conduct a spectrophotometric analysis to determine the concentration of $Cu^{2+}(aq)$ in a solution. The solution has a small amount of $Co(NO_3)_2(aq)$ present as a contaminant. The student is given the diagram below, which shows the absorbance curves for aqueous solutions of $Co^{2+}(aq)$ and $Cu^{2+}(aq)$.



(c) The spectrophotometer available to the student has a wavelength range of 400 nm to 700 nm. What wavelength should the student use to minimize the interference from the presence of the $Co^{2+}(aq)$ ions?

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8. Directions: For parts of the free-response question that require calculations, clearly show the method used and the steps involved in arriving at your answers. You must show your work to receive credit for your answer. Examples and equations may be included in your answers where appropriate.

Equimolar amounts of $F_2(g)$ and $O_2(g)$ are placed in a previously evacuated rigid container, where they react according to the equation below.

$$2F_2(g) + O_2(g) \rightleftarrows 2OF_2(g) \qquad \Delta H^{\circ}_{rxn} = +49.4\,\text{kJ/mol}_{rxn}$$

(a) If 3.8 g of $F_2(g)$ is consumed in the reaction, how many moles of $O_2(g)$ will be needed?

(b) Which element is reduced in this reaction? Justify your answer in terms of oxidation numbers.

The partial pressures of the gases in the container are monitored at constant temperature and recorded in the table below.

Time	P_{F_2} (atm)	P_{O_2} (atm)	P_{OF_2} (atm)
t_0	1.0	1.0	0.0
t_1	0.79	0.90	0.21
t_2	0.62	0.81	0.38
t_3	0.50	0.75	0.50
t_4	0.41	0.71	0.59
t_5	0.37	0.69	0.63
t_6	0.35	0.68	0.65
t_7	0.34	0.67	0.66
t_8	0.33	0.67	0.67
t_9	0.33	0.67	0.67

(c) At time t_2, is the rate of the forward reaction greater than, less than, or equal to the rate of the reverse reaction? Justify your choice.

(d) At equilibrium, the container holds fewer molecules of which gas, $F_2(g)$ or $O_2(g)$? Explain your answer.

(e) A student hypothesizes that if the temperature of the container is increased after time t_9, the mole fraction of OF_2 in the container will increase. Do you agree or disagree with the student's hypothesis? Justify your answer.

(f) Using the data in the table, determine the value of the equilibrium constant, K_p, for the reaction represented by the equation below.

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$$2\,F_2(g) + O_2(g) \rightleftarrows 2\,OF_2(g)$$

(g) Using the value determined in part (f), determine the value of $K_p\prime$ for the equation shown below.

$$2\,OF_2(g) \rightleftarrows 2\,F_2(g) + O_2(g) \qquad\qquad K_p\prime = \ ?$$

(h) Which gas, $F_2(g)$ or $OF_2(g)$, will behave most like an ideal gas at low temperature? Justify your choice.

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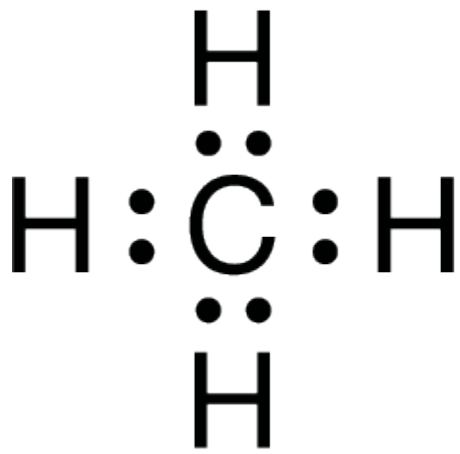
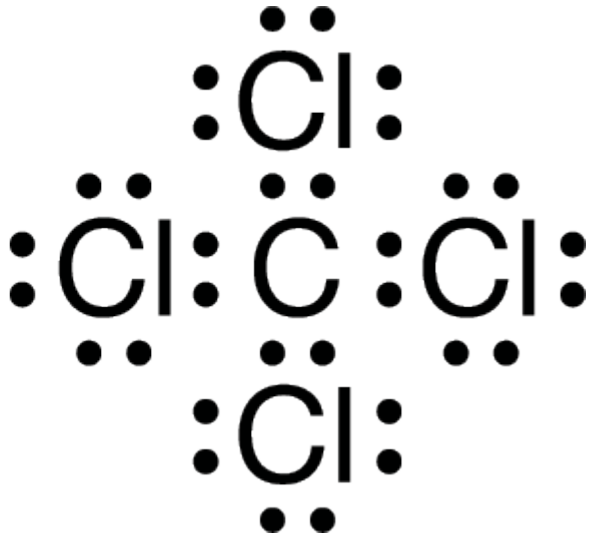
9. For parts of the free-response question that require calculations, clearly show the method used and the steps involved in arriving at your answers. You must show your work to receive credit for your answer. Examples and equations may be included in your answers where appropriate.





Answer the following questions about CH_4 and CCl_4. The Lewis electron-dot diagrams for molecules of the compounds are shown above.

(a) Which compound has the higher boiling point? Justify your answer. The type(s) of intermolecular force(s) in both substances should be included in your answer.

(b) At $400\ \text{K}$ both compounds are gases. At this temperature, which compound, $CH_4(g)$ or $CCl_4(g)$, behaves more like an ideal gas? Justify your answer, including reasoning about both molecules.

(c) $CCl_4(l)$ is placed in a previously evacuated container at 30°C, and some of the $CCl_4(l)$ evaporates. In the box below, draw a particulate diagram to show the species in the container after some of the $CCl_4(l)$ has evaporated. The phases of the species should be indicated by the spacing and distribution of the particles in the diagram. Some of the species shown in the legend will not be used.

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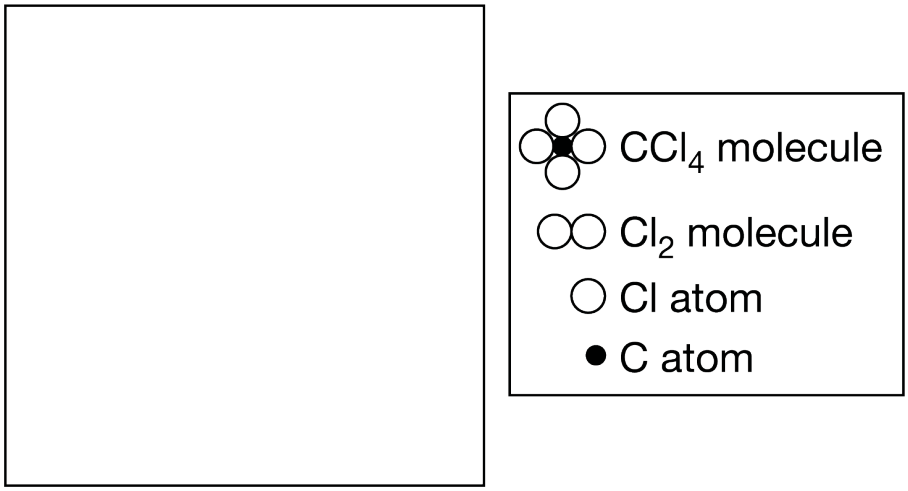

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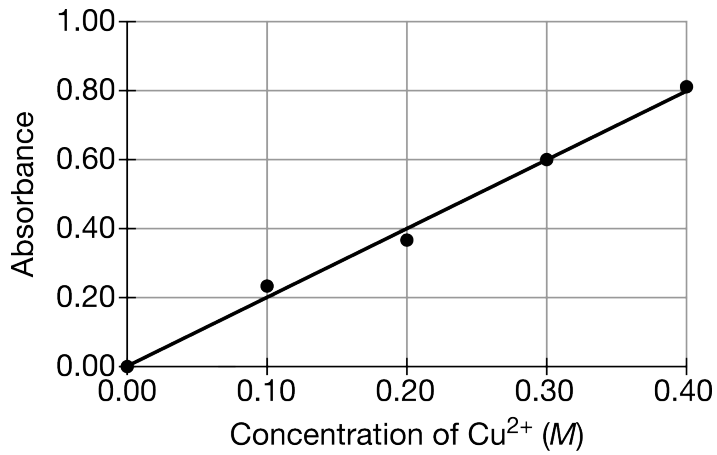
10. For parts of the free-response question that require calculations, clearly show the method used and the steps involved in arriving at your answers. You must show your work to receive credit for your answer. Examples and equations may be included in your answers where appropriate.

A student is given a sample of $CuSO_4(s)$ that contains a solid impurity that is soluble and colorless. The student wants to determine the amount of $CuSO_4$ in the sample and decides to use a spectrophotometer. First, the student prepares a calibration graph by measuring the absorbances of $CuSO_4(aq)$ solutions of known concentrations. The graph is shown below.

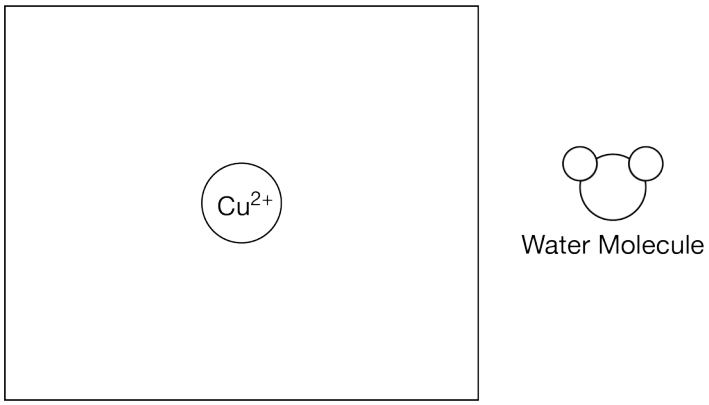


(a) The student dissolves the entire impure sample of $CuSO_4(s)$ in enough distilled water to make $100.\ mL$ of solution. Then the student measures the absorbance of the solution and observes that it is 0.30. Determine the concentration of $CuSO_4(aq)$ in the solution.

(b) Calculate the number of moles of $CuSO_4$ that were in the impure sample of $CuSO_4(s)$.

(c) In addition to the number of moles of $CuSO_4$ calculated in part (b), what other quantity must be measured in order to calculate the mass percentage of $CuSO_4$ in the impure sample of $CuSO_4(s)$?

(d) A particulate drawing of a single Cu^{2+} ion is shown in the following box. Draw <u>two</u> H_2O molecules in the box to show the most likely orientation of $H_2O(l)$ molecules around the Cu^{2+} ion.



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11. Directions: For parts of the free-response question that require calculations, clearly show the method used and the steps involved in arriving at your answers. You must show your work to receive credit for your answer. Examples and equations may be included in your answers where appropriate.

Equimolar amounts of $H_2(g)$ and $Br_2(g)$ are injected into an evacuated, rigid container, where they react according to the equation below.

$$H_2(g) + Br_2(g) \rightleftarrows 2\,HBr(g) \qquad \Delta H^{\circ}_{rxn} = -72.4\,\text{kJ/mol}_{rxn}$$

(a) If 1.6 g of Br_2 is consumed in the reaction with excess H_2, how many moles of HBr are produced?

(b) Which element is reduced in this reaction? Justify your answer in terms of the oxidation numbers.

The partial pressures of the gases in the container are monitored at constant temperature and recorded in the table below.

Time	P_{H_2} (atm)	P_{Br_2} (atm)	P_{HBr} (atm)
t_0	1.0	1.0	0.0
t_1	0.78	0.78	0.44
t_2	0.62	0.62	0.76
t_3	0.51	0.51	0.98
t_4	0.45	0.45	1.10
t_5	0.42	0.42	1.17
t_6	0.41	0.41	1.19
t_7	0.40	0.40	1.20
t_8	0.40	0.40	1.20
t_9	0.40	0.40	1.20

(c) At time t_9, is the rate of the forward reaction greater than, less than, or equal to the rate of the reverse reaction? Justify your choice.

(d) At equilibrium, the container holds the most molecules of which gas, H_2, Br_2, or HBr? Explain your answer.

(e) A student hypothesizes that if the temperature of the container is decreased after time t_9, the mole fraction of HBr in the container will decrease. Do you agree or disagree with the student's hypothesis? Justify your answer.

(f) Using the data in the table, determine the value of the equilibrium constant, K_p, for the reaction represented by the equation below.

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$$H_2(g) + Br_2(g) \rightleftarrows 2\,HBr(g)$$

(g) Using the value determined in part (f), determine the value of $K_p\prime$ for the equation shown below.

$$2\,H_2(g) + 2\,Br_2(g) \rightleftarrows 4\,HBr(g) \qquad\qquad K_p\prime = \ ?$$

(h) Which gas, $Br_2(g)$ or $H_2(g)$, will behave most like an ideal gas at low temperature? Justify your choice.

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12. For parts of the free response question that require calculations, clearly show the method used and the steps involved in arriving at your answers. You must show your work to receive credit for your answer. Examples and equations may be included in your answers where appropriate.

For parts of the free response question that require calculations, clearly show the method used and the steps involved in arriving at your answers. You must show your work to receive credit for your answer. Examples and equations may be included in your answers where appropriate.

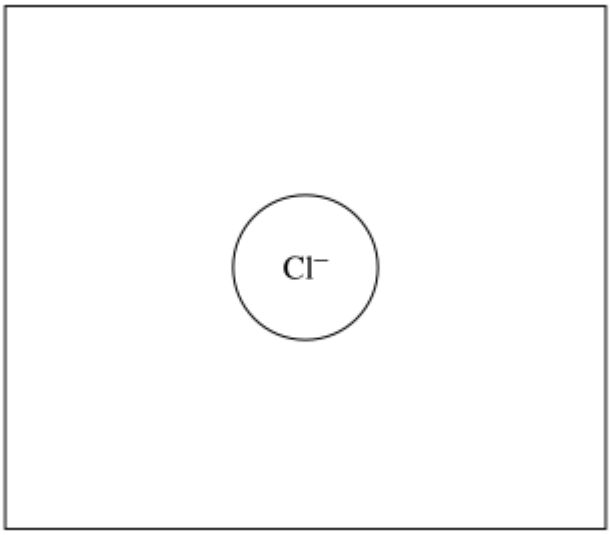
HCl is a molecular gas as a pure substance but acts as an acid in aqueous solution.

(a) A sample of $\mathrm{HCl}(g)$ is stored in a rigid $6.00\ \mathrm{L}$ container at $7.45\ \mathrm{atm}$ and $296\ \mathrm{K}$.

(i) Calculate the number of moles of $\mathrm{HCl}(g)$ in the container.

(ii) The rigid $6.00\ \mathrm{L}$ container of $\mathrm{HCl}(g)$ is cooled to a temperature of $271\ \mathrm{K}$. Calculate the new pressure, in atm, of the $\mathrm{HCl}(g)$.

(b) When HCl ionizes in aqueous solution, $\mathrm{Cl}^-(aq)$ ions are formed. In the following box, draw three water molecules with proper orientation around the Cl^- ion. Use to represent water molecules.



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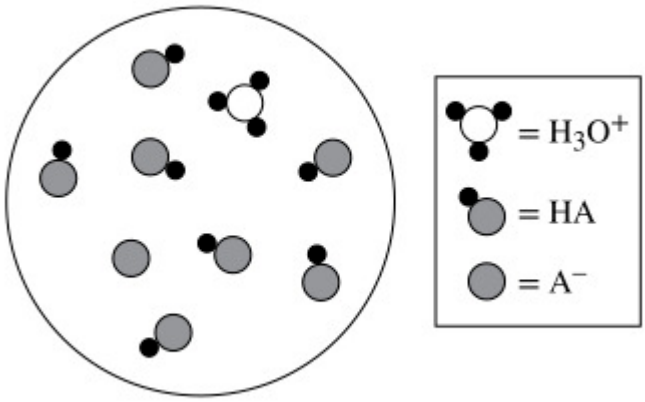
Acid (HA)	Anion (A^-)	K_a Value
HNO_2	NO_2^-	5.6×10^{-4}
HCl	Cl^-	2.0×10^{7}
$HClO_4$	ClO_4^-	1.6×10^{15}

The K_a values for three acids are shown in the preceding table.

(c) The following particulate diagram represents the ionization of one of the acids in the data table. Water molecules have been omitted for clarity. Which acid (HNO_2, HCl, or $HClO_4$) is represented in the diagram? Justify your answer using the information in the table.



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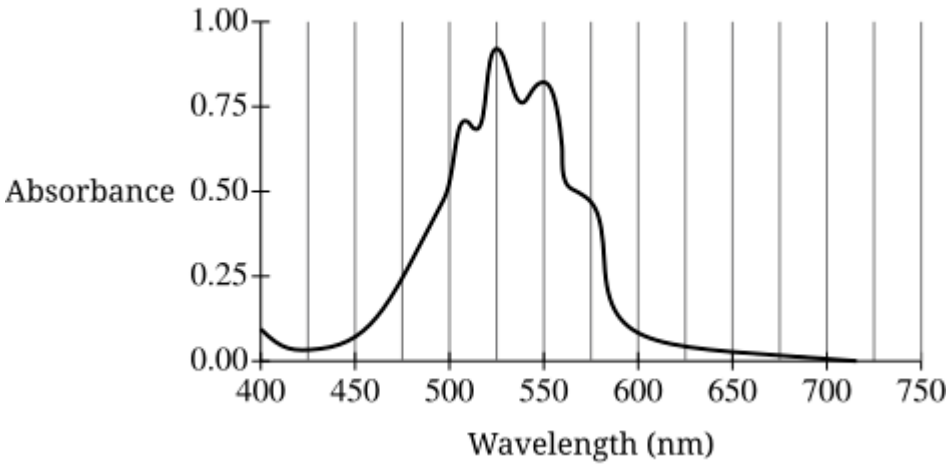
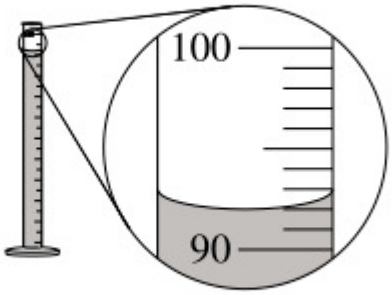
13. A student wants to determine the concentration of permanganate, $MnO_4^-(aq)$, in a solution. The student plans to use colorimetric analysis because solutions containing $MnO_4^-(aq)$ have a purple color.



(a) To determine the optimum wavelength for an experiment that measures the concentration of $MnO_4^-(aq)$, the student takes a sample of the solution and measures the amount of light absorbed by the sample over a range of wavelengths. The data are plotted in the graph shown. Identify the optimum wavelength that the student should use for the experimental procedure.

(b) The student uses a stock solution of $2.40 \times 10^{-3}\ M\ KMnO_4(aq)$ to prepare the standard solutions of $MnO_4^-(aq)$ that are needed to construct a calibration curve.



(i) The student uses a $100.0\ mL$ graduated cylinder to measure a certain volume of $KMnO_4(aq)$ stock solution, as shown in the diagram given. What volume should the student record?
(ii) Calculate the volume, in mL, of $2.40 \times 10^{-3}\ M\ KMnO_4(aq)$ that is required to produce $100.0\ mL$ of a standard $1.68 \times 10^{-3}\ M\ MnO_4^-(aq)$ solution.

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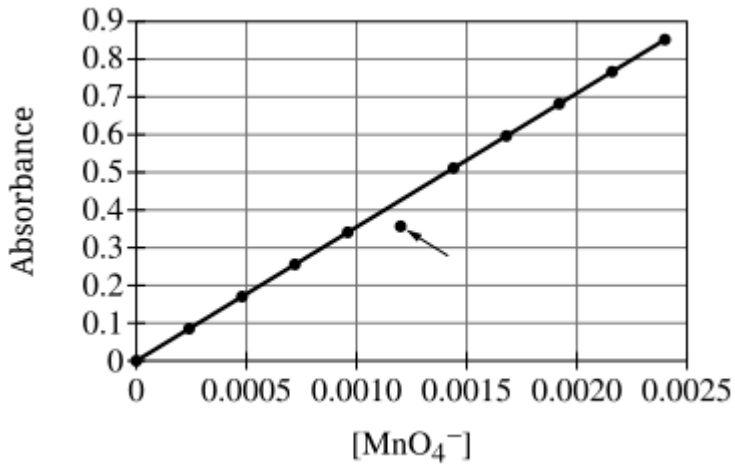
The student designs the following procedure to produce a calibration curve.

Step 1: Prepare several standard solutions that have known $MnO_4^-(aq)$ concentrations by dilution of the stock solution.
Step 2: Rinse the cuvette with distilled water.
Step 3: Rinse the cuvette with the standard solution and fill the cuvette with the standard solution.
Step 4: Measure the absorbance of the standard solution with the colorimeter.
Step 5: Repeat steps 2-4 for each of the standard solutions.

The data are plotted in the calibration curve shown. One of the data points (indicated with an arrow) on the calibration curve is below the line of best fit.



(c) Assuming that all lab equipment is functioning properly, identify which one of the procedural steps the student could have executed incorrectly that would explain why the marked data point is below the line of best fit. Justify your answer.

frq8

14. For parts of the free-response question that require calculations, clearly show the method used and the steps involved in arriving at your answers. You must show your work to receive credit for your answer. Examples and equations may be included in your answers where appropriate.

A student is doing experiments with $CO_2(g)$. Originally, a sample of the gas is in a rigid container at $299\ \mathrm{K}$ and $0.70\ \mathrm{atm}$. The student increases the temperature of the $CO_2(g)$ in the container to $425\ \mathrm{K}$.

(a) Describe the effect of raising the temperature on the motion of the $CO_2(g)$ molecules.

(b) Calculate the pressure of the $CO_2(g)$ in the container at $425\ \mathrm{K}$.

(c) In terms of kinetic molecular theory, briefly explain why the pressure of the $CO_2(g)$ in the container changes as it is heated to $425\ \mathrm{K}$.

(d) The student measures the actual pressure of the $CO_2(g)$ in the container at $425\ \mathrm{K}$ and observes that it is less than the pressure predicted by the ideal gas law. Explain this observation.

frq8

15. For parts of the free response question that require calculations, clearly show the method used and the steps involved in arriving at your answers. You must show your work to receive credit for your answer. Examples and equations may be included in your answers where appropriate.

For parts of the free response question that require calculations, clearly show the method used and the steps involved in arriving at your answers. You must show your work to receive credit for your answer. Examples and equations may be included in your answers where appropriate.

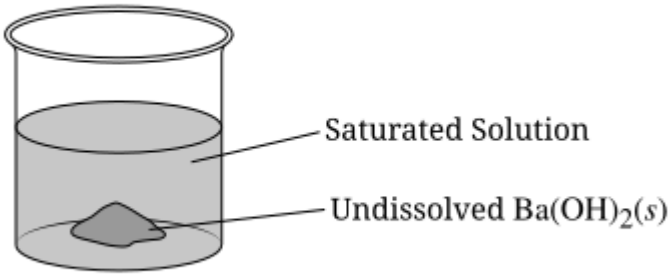
To determine the value of the solubility product constant, K_{sp}, a student prepares a saturated solution by dissolving $\mathrm{Ba(OH)_2}(s)$ in water. The K_{sp} expression is $K_{sp} = \left[\mathrm{Ba^{2+}}\right]\left[\mathrm{OH^-}\right]^2$.

(a) Write the balanced equation for the dissociation reaction that is consistent with the K_{sp} expression.

(b) The student determines that the solubility of $\mathrm{Ba(OH)_2}(s)$ is $18.5 \quad \mathrm{g/L}$ at $25^\circ\mathrm{C}$.

(i) Calculate the molar solubility of $\mathrm{Ba(OH)_2}(s)$ in $\mathrm{mol/L}$ at $25^\circ\mathrm{C}$.

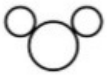
(ii) Calculate the value of K_{sp} for $\mathrm{Ba(OH)_2}(s)$ at $25^\circ\mathrm{C}$.



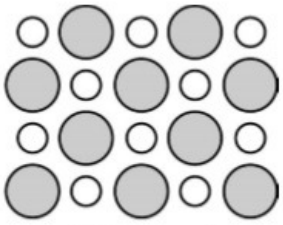
(c) The student creates a saturated solution of $\mathrm{Ba(OH)_2}$ in water, as shown in the diagram. The student claims that the amount of undissolved $\mathrm{Ba(OH)_2}(s)$ will decrease when $\mathrm{HNO_3}(aq)$ is added. Do you agree or disagree with the student's claim? Justify your answer with a comparison of Q and K_{sp}.

frq8

16.



H_2O molecule



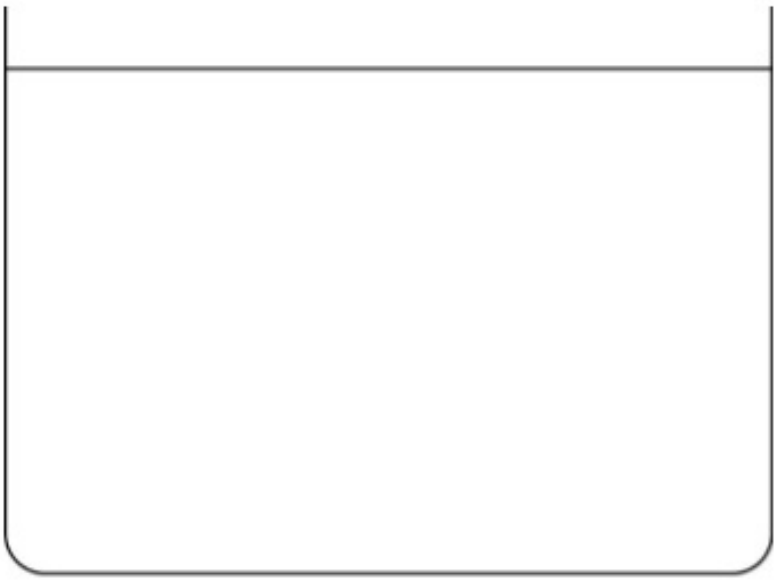
LiCl crystal

The structures of a water molecule and a crystal of LiCl(*s*) are represented above. A student prepares a 1.0 *M* solution by dissolving 4.2 g of LiCl(*s*) in enough water to make 100 mL of solution.

a. In the space provided below, show the interactions of the components of LiCl(*aq*) by making a drawing that represents the different particles present in the solution. Base the particles in your drawing on the particles shown in the representations above. Include only one formula unit of LiCl and no more than ten molecules of water. Your drawing must include the following details.

• identity of ions (symbol and charge)

• the arrangement and proper orientation of the particles in the solution



LiCl (*aq*)

b. The student passes a direct current through the solution and observes that chlorine gas is produced at the anode. Identify the chemical species produced at the cathode and justify your answer using the information given in the table below.

frq8

Half-reaction	Standard Reduction Potential at 25°C (V)
$Li^+(aq) + e^- \rightarrow Li(s)$	− 3.05
$2\ H_2O(l) + 2\ e^- \rightarrow H_2(g) + 2\ OH^-(aq)$	− 0.83

frq8

17. Directions: For parts of the free-response question that require calculations, clearly show the method used and the steps involved in arriving at your answers. You must show your work to receive credit for your answer. Examples and equations may be included in your answers where appropriate.

Equimolar amounts of $NO(g)$ and $O_2(g)$ are injected into an evacuated, rigid container, where they react according to the equation below.

$$2\ NO(g) + O_2(g) \rightleftarrows 2\ NO_2(g) \qquad \Delta H^{\circ}_{rxn} = -112\,\text{kJ/mol}_{rxn}$$

(a) If 3.2 g of $O_2(g)$ is consumed in the reaction with excess $NO(g)$, how many moles of $NO_2(g)$ are produced?

(b) Which element is oxidized in this reaction? Justify your answer in terms of oxidation numbers.

The partial pressures of the gases in the container are monitored at constant temperature and recorded in the table below.

Time	P_{NO} (atm)	P_{O_2} (atm)	P_{NO_2} (atm)
t_0	1.0	1.0	0.0
t_1	0.80	0.90	0.20
t_2	0.69	0.84	0.31
t_3	0.62	0.81	0.38
t_4	0.58	0.79	0.42
t_5	0.55	0.77	0.45
t_6	0.52	0.76	0.48
t_7	0.50	0.75	0.50
t_8	0.50	0.75	0.50
t_9	0.50	0.75	0.50

(c) At time t_3, is the rate of the forward reaction greater than, less than, or equal to the rate of the reverse reaction? Justify your choice.

(d) At equilibrium, the container holds the most molecules of which gas, NO, O_2, or NO_2? Explain your answer.

frq8

(e) A student hypothesizes that if the temperature of the container is increased after time t_9, the mole fraction of O_2 in the container will increase. Do you agree or disagree with the student's hypothesis? Justify your answer.

(f) Using the data in the table, determine the value of the equilibrium constant, K_p, for the reaction represented by the equation below.

$$2\ NO(g) + O_2(g) \rightleftarrows 2\ NO_2(g)$$

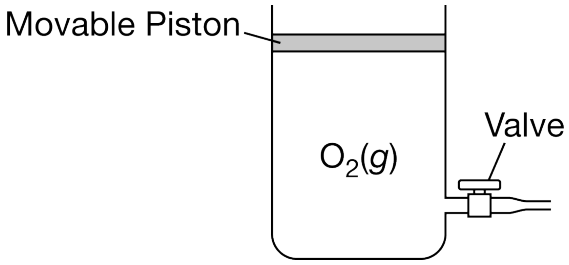
(g) Using the value determined in part (f), determine the value of $K_p\prime$ for the equation shown below.

$$2\ NO_2(g) \rightleftarrows 2\ NO(g) + O_2(g) \qquad K_p\prime = \ ?$$

(h) Which gas, $NO_2(g)$ or $O_2(g)$, will behave most like an ideal gas at low temperature? Justify your choice.

frq8

18. For parts of the free response question that require calculations, clearly show the method used and the steps involved in arriving at your answers. You must show your work to receive credit for your answer. Examples and equations may be included in your answers where appropriate.



A student investigates gas behavior using a rigid cylinder with a movable piston of negligible mass, as shown in the diagram above. The cylinder contains $0.325\ \text{mol}$ of $O_2(g)$.

(a) The cylinder has a volume of $7.95\ \text{L}$ at 25°C and $1.00\ \text{atm}$. Calculate the density of the $O_2(g)$, in g/L, under these conditions.

(b) Attempting to change the density of the $O_2(g)$, the student opens the valve on the side of the cylinder, pushes down on the piston to release some of the gas, and closes the valve again. The temperature of the gas remains constant at 25°C. Will this action change the density of the gas remaining in the cylinder? Justify your answer.

(c) The student tries to change the density of the $O_2(g)$ by cooling the cylinder to -55°C, which causes the volume of the gas to decrease. Using principles of kinetic molecular theory, explain why the volume of the $O_2(g)$ decreases when the temperature decreases to -55°C.

(d) The student further cools the cylinder to -180°C and observes that the measured volume of the $O_2(g)$ is substantially smaller than the volume that is calculated using the ideal gas law. Assume all equipment is functioning properly. Explain why the measured volume of the $O_2(g)$ is smaller than the calculated volume. (The boiling point of $O_2(l)$ is -183°C.)

frq8

19. For parts of the free-response question that require calculations, clearly show the method used and the steps involved in arriving at your answers. You must show your work to receive credit for your answer. Examples and equations may be included in your answers where appropriate.

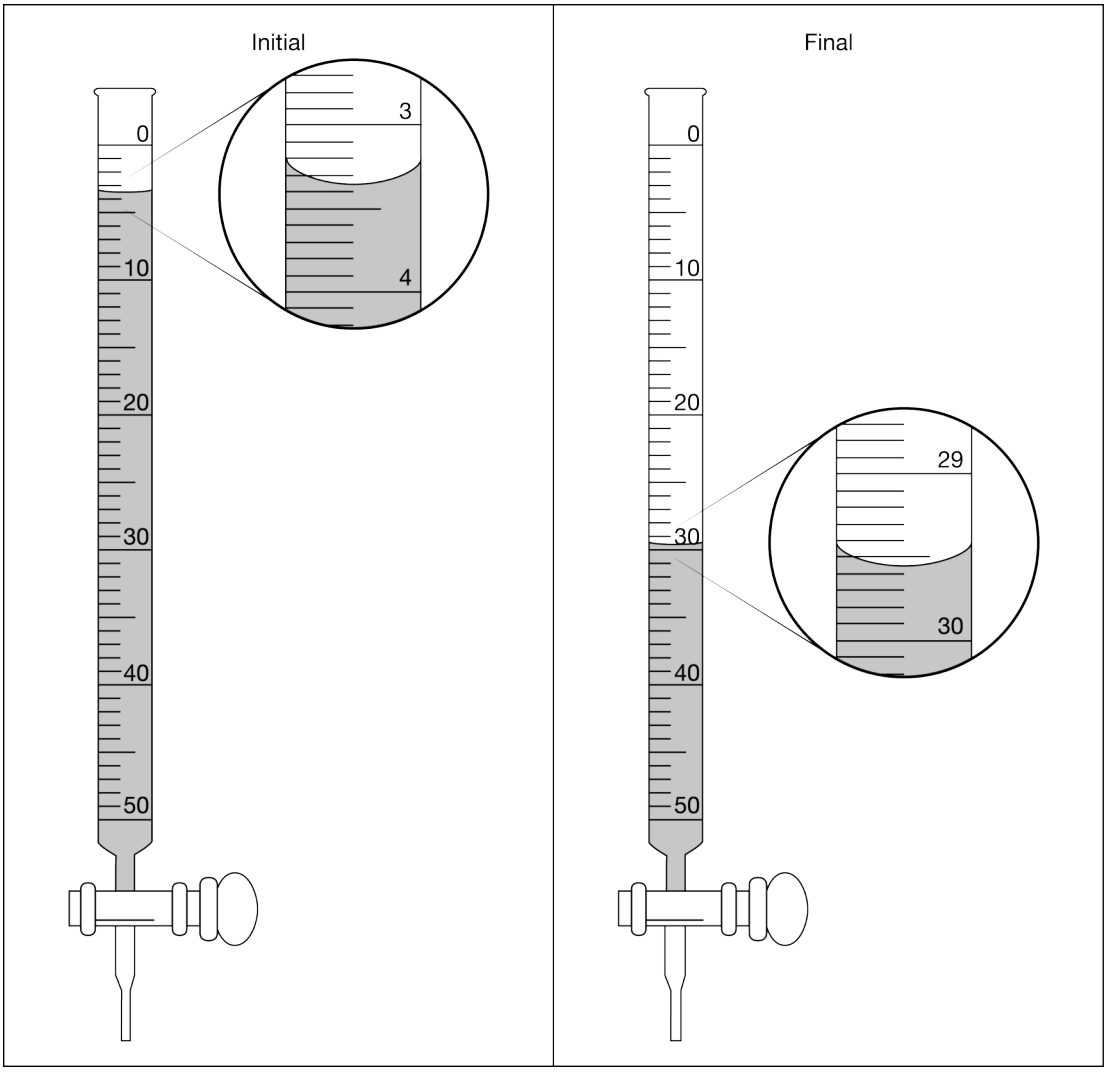
$$6\ H^+(aq) + 2\ MnO_4^-(aq) + 5\ H_2C_2O_4(aq) \rightarrow 10\ CO_2(g) + 8\ H_2O(l) + 2\ Mn^{2+}(aq)$$

A student dissolved a $0.139\ g$ sample of oxalic acid, $H_2C_2O_4$, in water in an Erlenmeyer flask. Then the student titrated the $H_2C_2O_4$ solution in the flask with a solution of $KMnO_4$, which has a dark purple color. The balanced chemical equation for the reaction that occurred during the titration is shown above.

(a) Identify the species that was reduced in the titration reaction. Justify your answer in terms of oxidation numbers.

(b) The student used a $50.0\ mL$ buret to add the $KMnO_4(aq)$ to the $H_2C_2O_4(aq)$ until a faint lavender color was observed in the flask, an indication that the end point of the titration had been reached. The initial and final volume readings of the solution in the buret are shown below. Write down the initial reading and the final reading and use them to determine the volume of $KMnO_4(aq)$ that was added during the titration.

frq8



(c) Given that the concentration of $KMnO_4(aq)$ was $0.0235\ M$, calculate the number of moles of MnO_4^- ions that completely reacted with the $H_2C_2O_4$.

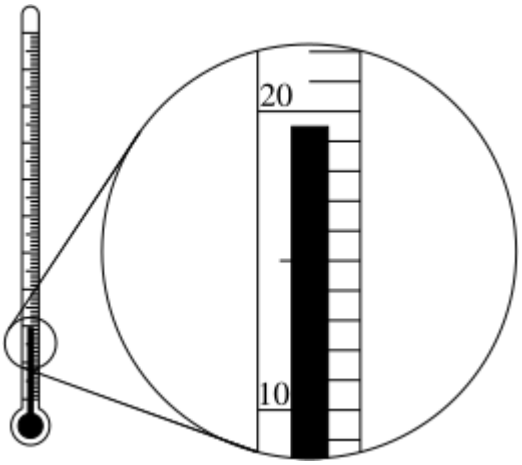
(d) The student proposes to perform another titration using a 0.139 g sample of $H_2C_2O_4$, but this time using $0.00143\ M\ KMnO_4(aq)$ in the buret. Would this titrant concentration be a reasonable choice to use if the student followed the same procedure and used the same equipment as before? Justify your response.

frq8

20. For parts of the free response question that require calculations, clearly show the method used and the steps involved in arriving at your answers. You must show your work to receive credit for your answer. Examples and equations may be included in your answers where appropriate.

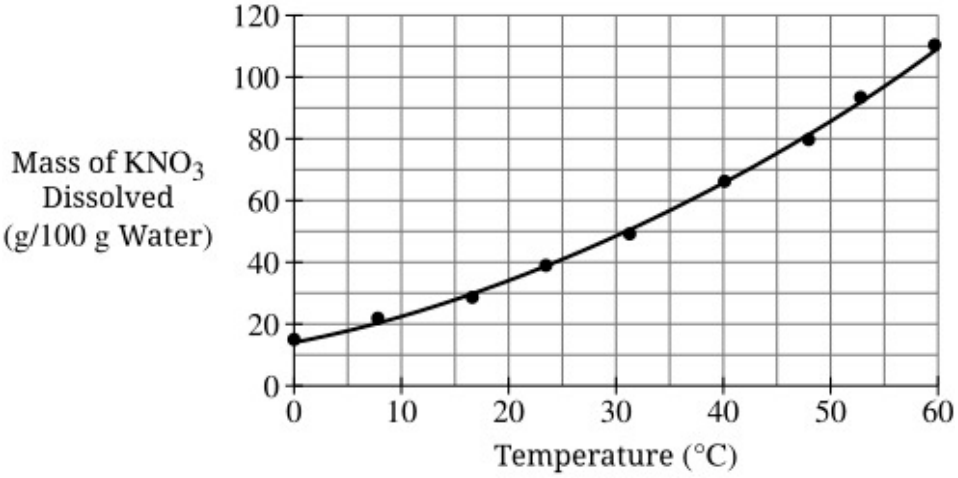
A student conducts a solubility experiment with potassium nitrate, KNO_3, in water by identifying the temperature at which various masses of KNO_3 will completely dissolve.

(a) At room temperature, H_2O is a liquid and KNO_3 is a solid. The student incorrectly claims that this is because potassium and nitrate ions experience stronger dispersion forces than H_2O molecules. In terms of the types of attractive forces, explain why the student is incorrect.



(b) In one of the trials, the student reads the initial temperature of the water using the Celsius thermometer above. Record this temperature to the appropriate number of significant figures.

(c) At the temperature indicated on the thermometer, the student adds the solid KNO_3 to the sample of water and begins to stir the mixture. Explain the effect of stirring on the maximum amount of solid that can dissolve in the water.



frq8

The graph above summarizes the data collected by the student.

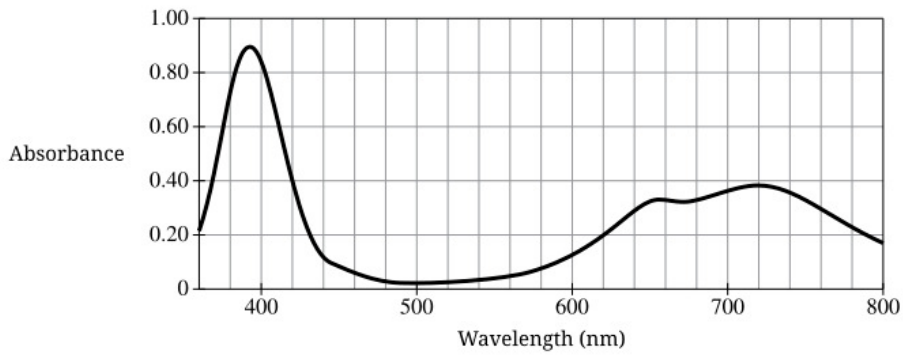
(d) Another student places 38.1 grams of KNO_3 into 100.0 g of water at the temperature measured in part (b), stirs the mixture, and observes that the solid does not dissolve completely. The student plans to heat the mixture and states that the solid will completely dissolve before the mixture reaches $30\,^{\circ}C$. Do you agree with the claim? Justify your answer with information from the graph.

frq8

21. For parts of the free response question that require calculations, clearly show the method used and the steps involved in arriving at your answers. You must show your work to receive credit for your answer. Examples and equations may be included in your answers where appropriate.

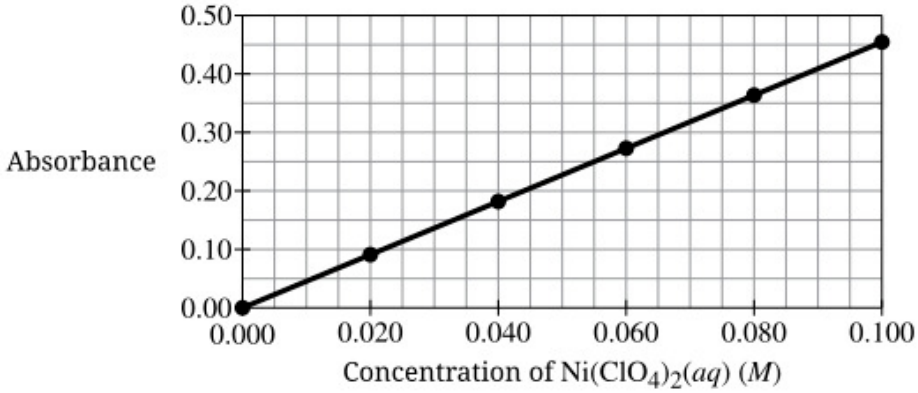
For parts of the free response question that require calculations, clearly show the method used and the steps involved in arriving at your answers. You must show your work to receive credit for your answer. Examples and equations may be included in your answers where appropriate.

A student performs a spectrophotometry experiment using solutions of $\mathrm{Ni}\,(\mathrm{ClO_4})_2(aq)$. The student needs to make a standard curve and determine the concentration of $\mathrm{Ni}\,(\mathrm{ClO_4})_2(aq)$ in a sample with an unknown concentration.



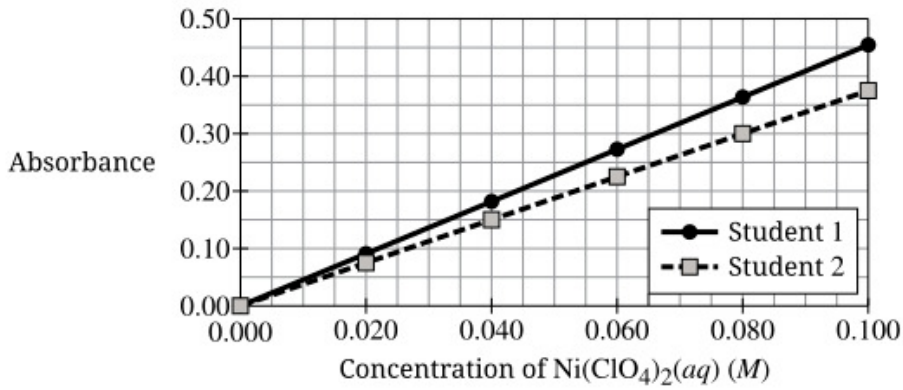
(a) Based on the absorbance spectrum of $\mathrm{Ni}\,(\mathrm{ClO_4})_2(aq)$ shown in the preceding graph, what wavelength should the student use in the experiment to achieve the greatest precision?

(b) To make the standard curve, the student must prepare $\mathrm{Ni}\,(\mathrm{ClO_4})_2(aq)$ solutions with different concentrations. The student has a standard solution of $\mathrm{Ni}\,(\mathrm{ClO_4})_2(aq)$ with a concentration of $0.198\ M$. Calculate the volume, in mL, of $0.198\ M\quad \mathrm{Ni}\,(\mathrm{ClO_4})_2(aq)$ that is needed to prepare $250.0\ \mathrm{mL}$ of $0.0400\ M\quad \mathrm{Ni}\,(\mathrm{ClO_4})_2(aq)$.



frq8

(c) The standard curve generated by the student is provided above. The student measures an absorbance of 0.30 for the unknown solution. Using the standard curve provided, determine the molarity of $\mathrm{Ni}\left(\mathrm{ClO}_4\right)_2(aq)$ in the unknown solution.



(d) A second student repeats the procedure using the same spectrophotometer but makes an error, which results in the dashed line shown in the graph above. Assuming that the first student's data (shown by the solid line) are correct, identify one error that could account for the second student's standard curve being less steep than that obtained by the first student.

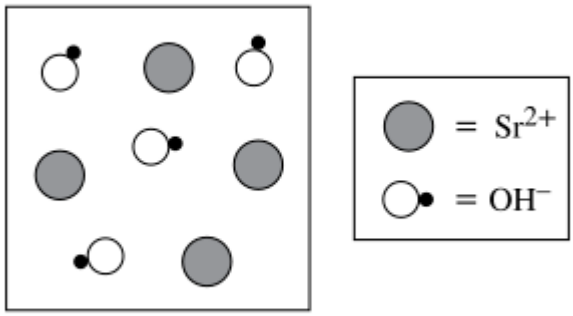
frq8

22. For parts of the free response question that require calculations, clearly show the method used and the steps involved in arriving at your answers. You must show your work to receive credit for your answer. Examples and equations may be included in your answers where appropriate.

For parts of the free response question that require calculations, clearly show the method used and the steps involved in arriving at your answers. You must show your work to receive credit for your answer. Examples and equations may be included in your answers where appropriate.

Strontium hydroxide dissolves in water according to the following equation. The K_{sp} expression for strontium hydroxide is provided.

$$\mathrm{Sr(OH)_2}(s) \rightleftarrows \mathrm{Sr^{2+}}(aq) + 2\ \mathrm{OH^-}(aq) \qquad K_{sp} = \left[\mathrm{Sr^{2+}}\right]\left[\mathrm{OH^-}\right]^2$$



(a) A student draws the particulate diagram shown to represent the ions present in an aqueous solution of $\mathrm{Sr(OH)_2}$. (Water molecules are intentionally omitted.) Identify the error in the student's drawing.

(b) The student prepares a saturated solution by adding excess $\mathrm{Sr(OH)_2}(s)$ to distilled water and stirring until no more solid dissolves. The student then determines that $\left[\mathrm{Sr^{2+}}\right] = 0.043\ M$ in the solution.

(i) Calculate the value of $\left[\mathrm{OH^-}\right]$ in the solution.

(ii) Calculate the value of K_{sp} for $\mathrm{Sr(OH)_2}$.

(c) The student prepares a second saturated solution of $\mathrm{Sr(OH)_2}$ in aqueous $0.10\ M\ \mathrm{Sr(NO_3)_2}$ instead of water. Will the value of $\left[\mathrm{OH^-}\right]$ in the second solution be greater than, less than, or equal to the value in the first solution? Justify your answer. (Assume constant temperature.)